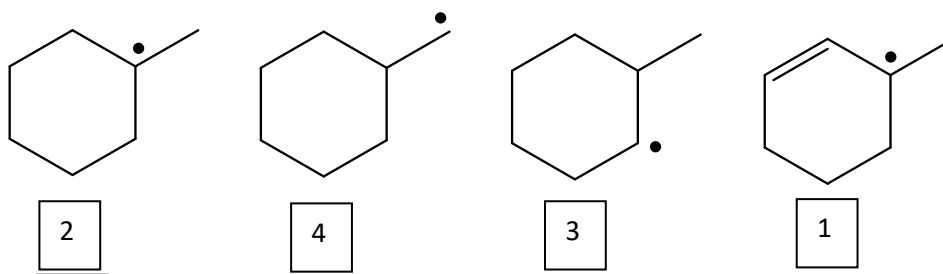


CH 221

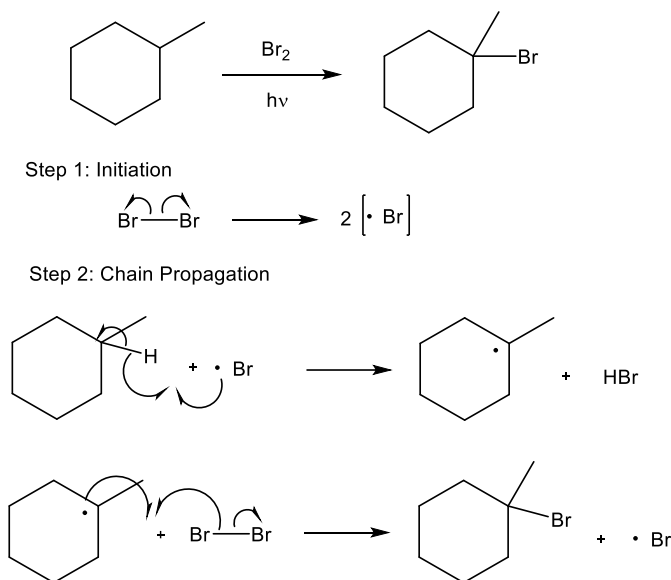
CH 10 Worksheet KEY: Radical Reactions

1. Rank the following radicals in order of stability (1 = most stable, 4 = least stable).



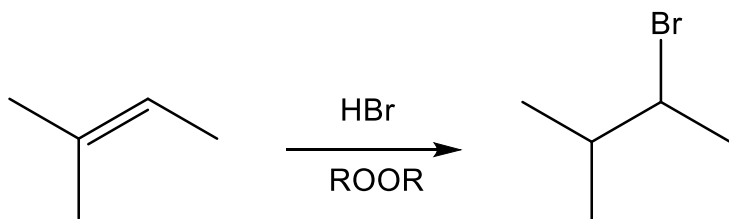
2. Halogenation of alkanes generally proceed by way of free-radical intermediates during a photochemical reaction. What are the three steps in a radical reaction chain mechanism? Draw out the step-by-step reaction mechanism for the reaction below.

*Three steps are initiation, chain propagation and termination.

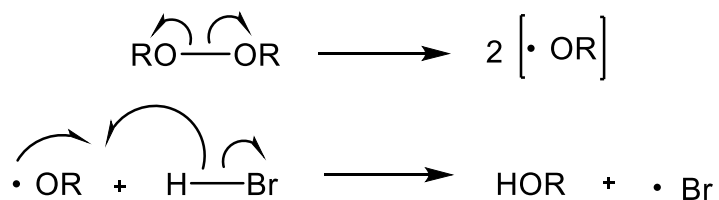


*Chain propagation repeat many times. Then, termination can occur between two bromine atoms, two alkyl radicals, and/or alkyl radical with bromine atom.

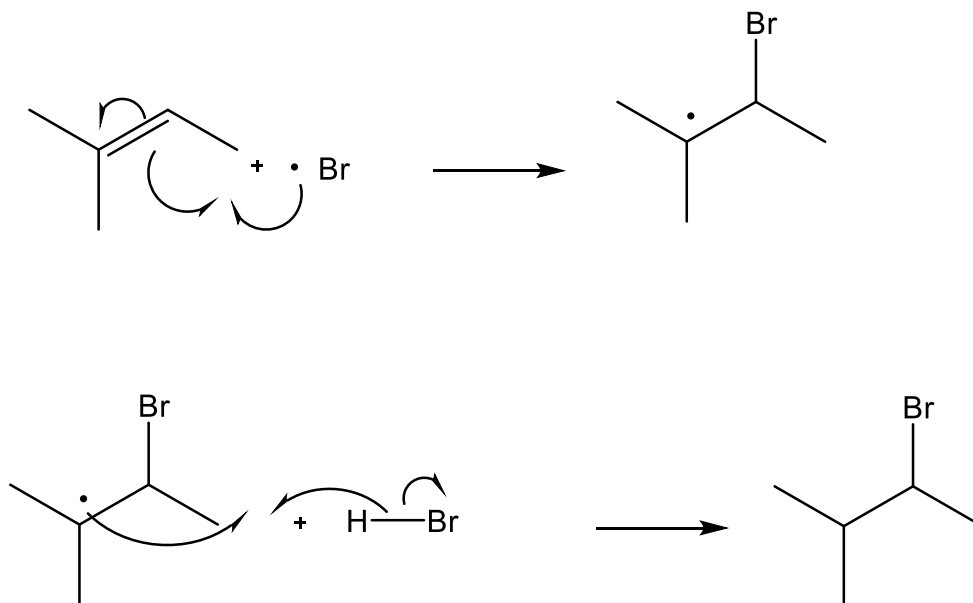
3. Draw the mechanism and the major product(s) for the following reaction.



Step 1: Initiation



Step 2: Chain Propagation

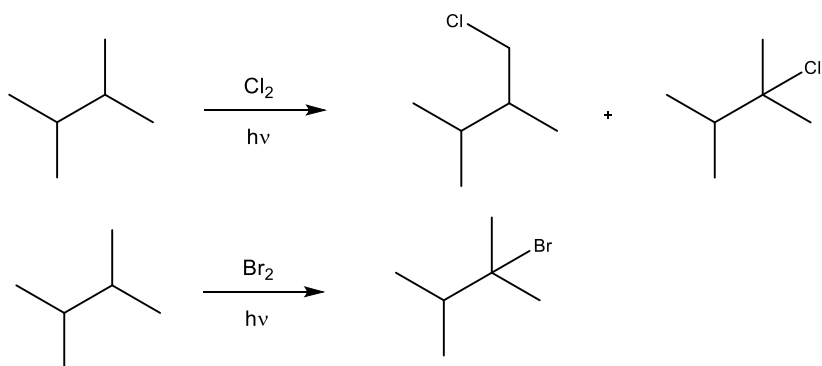


*Chain propagation repeat many times. Then, termination can occur between two bromine atoms, two alkyl radicals, and/or alkyl radical with bromine atom.

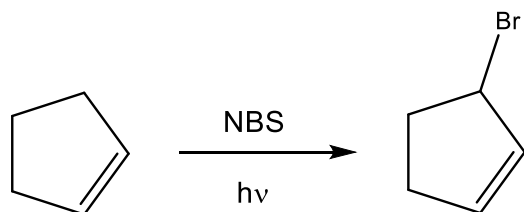
4. A. What is the difference in reactivity between a bromine radical and a chlorine radical?

A bromine radical is more stable than a chlorine radical, therefore when an alkane photochemically reacts with chlorine, it is not very selective. The photochemical reaction between bromine and an alkane is much more selective.

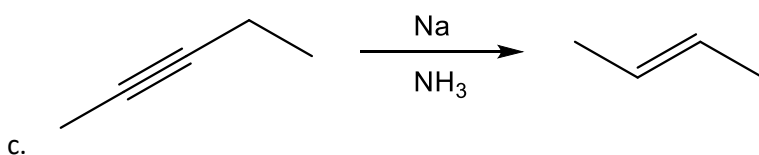
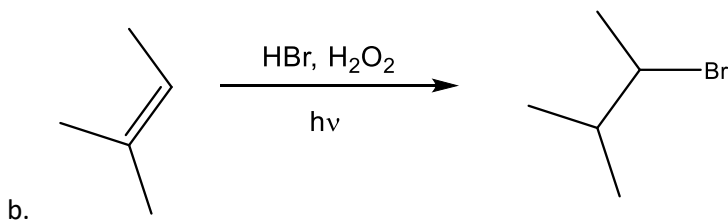
B. Determine the major product (s) for the following reactions.



5. Provide the product for each of the following reactions.



a. **NBS (N-Bromosuccinimide) is a reagent that is a bromine source for allylic bromination.



6. The following involve multistep synthesis. Propose a synthesis for the following:

